

We were discussing the protection problem of the LAA, and the idea was to minimise the sum of all P_v^{LP} (load to be protected). This is a minimum effort protection.

Let's get back to the stability constraint of the problem. We've seen how we need to define a Lyapunov criterion for asymptotic stability.

We've seen how we need to find a $V : \mathbb{R}^n \rightarrow \mathbb{R}$ for x^{eq} to be asymptotically stable such that $V > 0$ and $\dot{V} < 0$.

What is $\dot{V}(x(t)) = \frac{dV}{dx} \cdot \frac{dx}{dt}$, where $\frac{dx}{dt}$ is the model of the system ($\dot{x} = f(x)$), and $\frac{dV}{dx}$ is the gradient of V with respect to x .

Example:

$$x \in \mathbb{R}^2 \quad \begin{cases} \dot{x}_1 = x_1 + x_2 \\ \dot{x}_2 = \sin(x_1) \end{cases}$$

We take $V = 2x_1^2 + x_2^2$.

Let's focus on $x^{eq} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$. The function is > 0 in x^{eq} , as $V(0) = 0$.

This is globally positive-definite.

Let's compute now $\dot{V}$.

$$\begin{aligned} \dot{V} &= \frac{dV}{dx} \cdot \frac{dx}{dt} \\ \frac{dV}{dx} &= (4x_1 \quad 2x_2) \\ \dot{V} &= (4x_1 \quad 2x_2) \cdot \begin{pmatrix} x_1 + x_2 \\ \sin(x_1) \end{pmatrix} = 4x_1^2 + 4x_1x_2 + 2\sin(x_1)x_2 \end{aligned}$$

We could check, and this function technically is not negative definite.

Let's see now Lyapunov's theory for linear systems.

Consider a system $\dot{x} = Ax$. We have that $x^{eq} = 0$.

Consider a particular positive-definite function, a quadratic form. It's something like this: $V(x) = x^T \mathcal{X} x$, with $\mathcal{X}$ an $n \times n$ matrix, positive-definite.

A trivial example could be given by:

$$V(x) = \begin{pmatrix} x_1 & x_2 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 0 & 5 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} > 0$$

Now, let's see $\dot{V}$ calculation.

$$\dot{V} = \frac{dV}{dx} \cdot \dot{x} \quad \text{with } V = x^T \mathcal{X} x \text{ and } \dot{x} = Ax$$

We have the following known formula:

$$\frac{dV}{dx} = \frac{d(x^T \mathcal{X} x)}{dx} = 2x^T \mathcal{X}$$

and so

$$\dot{V} = 2x^T \mathcal{X} Ax = x^T \mathcal{X} Ax + x^T \mathcal{X} Ax =$$

These values are numbers, and because a number is equal to its transpose we get

$$= x^T \mathcal{X} Ax + x^T A^T \mathcal{X}^T x =$$

then, let's consider to take $\mathcal{X}$ symmetric, so that $\mathcal{X}^T = \mathcal{X}$, then

$$= x^T \mathcal{X} Ax + x^T A^T \mathcal{X} x$$

we gather x^T on the left and x on the right

$$\dot{V} = x^T \underbrace{(\mathcal{X}A + A^T \mathcal{X})}_{n \times n \text{ matrix, } P} x < 0$$

$$\dot{V} = x^T P x < 0$$

We thus have that $\dot{V}$ is a quadratic form, and the associated P must be negative-definite, $P < 0$, or $(\mathcal{X}A + A^T \mathcal{X}) < 0$.

To summarise: $x^{eq} = 0$ of $\dot{x} = Ax$ is A.S. if we can find $\mathcal{X} > 0$ such that $(\mathcal{X}A + A^T \mathcal{X}) < 0$. This is a matrix inequality.

Thus, the final protection problem (against multi-point attacks) is given by:

$$\begin{aligned} & \min_{(P^{LP}, K^{LG}, x)} \sum_v P_v^{LP} \\ & \text{subject to} \\ & P_v^{LP} \leq P_v^{LV} \quad \forall v \\ & 2\omega_s^{max} K_v^{LG} = (P_v^{LV} - P_v^{LP}) \quad \forall v \\ & \mathcal{X} > 0 \quad *_1 \\ & \mathcal{X}(A + B \begin{bmatrix} 0 & 0 & K^{LG} \end{bmatrix}) + (\dots)^T \mathcal{X} < 0 \quad *_2 \end{aligned}$$

We have that $*_1$ and $*_2$ are matrix inequalities to guarantee stability.

Given $P^{LP} = 0$ and $K^{LG} = 0$, we have a trivial solution.

Zero Dynamics Attacks

Reference scenario is given by the usual feedback control system, with the process sending measurements to a controller through a network. The controller computes the control, which goes to the plant. We have a monitor/detector which takes both y_k and u_k , and output the residual r_k .

The assumptions are:

- (P, C, D) are linear systems.
- Discrete time.

We have an attacker in the middle as usual, and he can inject something in the measurements or in the controls. In the work, however, we just consider manipulations in the control inputs.

What is the attack goal? It's to compute an attack sequence that:

1. Will have a big impact on the internal state of P , x_k . For example, some state variables go outside of safety limits.
2. Very small (or ideally, no) impact on the measured output y_k . When we talk about "no impact" we're saying how the attack is undetectable.

There are many way/techniques to find attack sequences achieving the two goals. We're going to see one way based on invariance.

Let's do modelling of the system now.

Starting from the process under attack:

$$\begin{cases} x_{k+1} = Ax_k + Bu_k + Ba_k \\ y_k = Cx_k \end{cases}$$

We neglect Du_k for y_k .

For the controller, we write:

$$u_k = \mathcal{F}(y_k)$$

So, the control is a linear function of the measurements.

For the detector, we write

$$r_k = \mathcal{D}(y_k, u_k)$$

How to analyse the attack?

We follow a method based on decomposition of variables. For the state variables,

we have that x_k is the state of the system during the attack, and we decompose this in two terms: $x_k = \bar{x}_k + x_k^a$ where $\bar{x}_k$ is the nominal evolution of the system (how the system evolves with no attack) and x_k^a is the impact, effect of the attack on the state.

When we start the attack (time k_0), the effect of the injection will be seen one step later.

x_k^a is the key variable to model the impact of the attack. Let's see how x_k^a varies depending on a_k . We want to find a dynamical equation of x_k^a , with x_{k+1}^a .

We have that:

$$\begin{aligned} x_{k+1}^a &= x_{k+1} - \bar{x}_{k+1} = Ax_k + Bu_k + Ba_k + (-A\bar{x}_k + B\bar{u}_k) = \\ &= A(x_k - \bar{x}_k) + B(u_k - \bar{u}_k) + Ba_k \end{aligned}$$

We have that $x_k^a = x_k - \bar{x}_k$, and $u_k^a = (u_k - \bar{u}_k)$, and we get that

$$x_{k+1}^a = Ax_k^a + Bu_k^a + Ba_k$$

Let's now compute y_k^a .

$$y_k^a = y_k - \bar{y}_k = Cx_k - C\bar{x}_k = Cx_k^a$$

By gathering these two equations we get the "delta" system, modelling the impact of attack on x and y .

$$\begin{cases} x_{k+1}^a = Ax_k^a + Bu_k^a + Ba_k \\ y_k^a = Cx_k^a \end{cases}$$

with the initial condition that $x_{k_0}^a = 0$.